

Diploma Thesis

Investigation of Machine Vibration Analysis System Design

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- ② Background
- ③ Engineering and Design
- ④ Implementation
- ⑤ Results
- ⑥ Conclusions

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Predictive Maintenance and Condition Monitoring

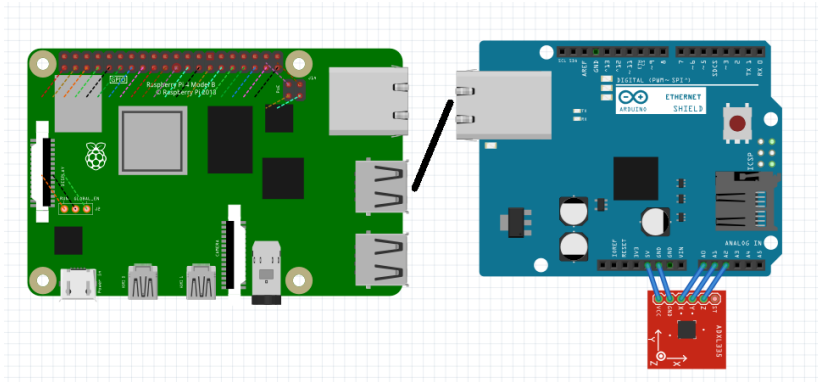


Figure 1: Hardware architecture with signal flow.

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- Development of a React frontend that requests and visualizes data from the Vercel API.

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Equipment Operational Anomalies

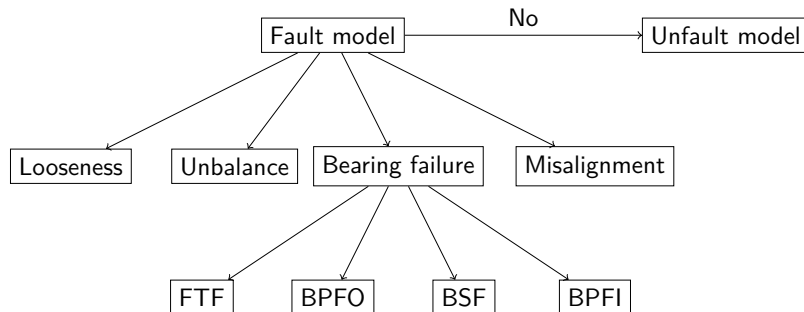


Figure 2: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

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System Architecture and Dataflow

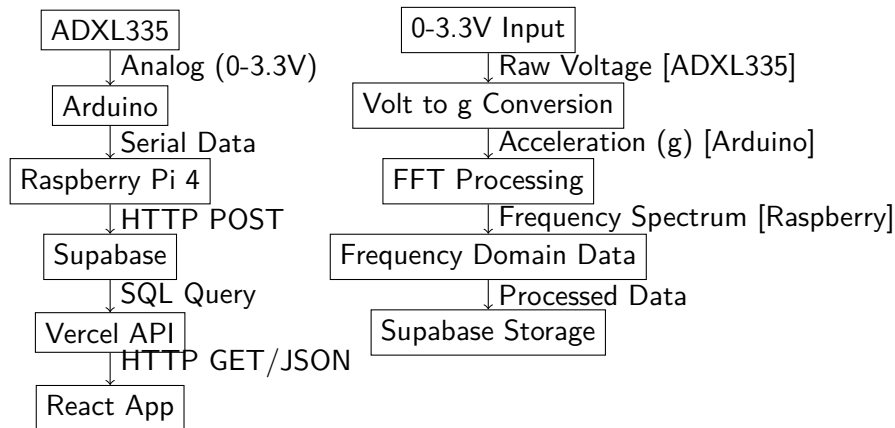


Figure 3: Complete system architecture and data flow and transformation pipeline from physical sensor to web visualization dashboard.

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Fault Type Classification

Power Spectrum Density calculation

Dynamic signal stability analysis

Vibration Spectrum and phase analysis

Incipient SVM on time-domain

Incipient fault detection with one-class v-SVM algorithm

Real-time fault detection set-up with SVM algorithm

Backpropagation Neural Network

Fault Type Classification

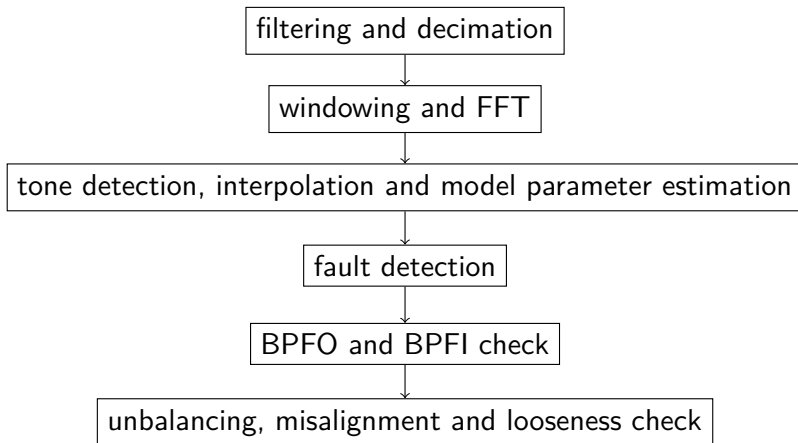


Figure 4: The series of checks carried out by the algorithm

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Power Spectrum Density calculation

Power Spectrum Density Threshold Calculation:

$$\text{PSD} = \frac{(\text{Power Spectrum})^2}{\Delta f \times \text{Noise Power Bandwidth of Window}}$$

Where:

- Power spectrum is calculated by taking the average of the magnitude squared of multiple frequency spectra

$$PS = \frac{2 \times X^2}{N^2}$$

-

$$\Delta f = \frac{f_s}{N}$$

is the frequency resolution

- Noise Power Bandwidth of Window is a correction factor for the rectangular window

Power Spectrum Density calculation

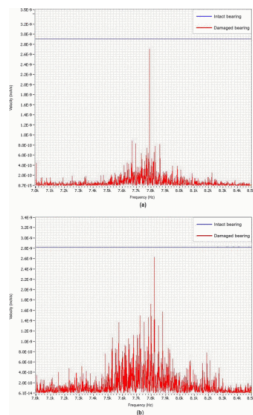


Figure 5: Combination of all the aforementioned faults (inner case, outer case and bearing ball fault) [?].

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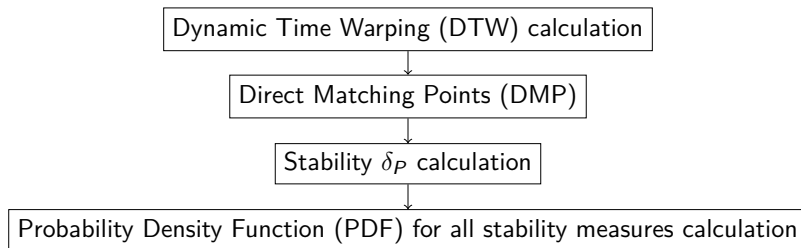


Figure 6: The series of checks carried out by the algorithm

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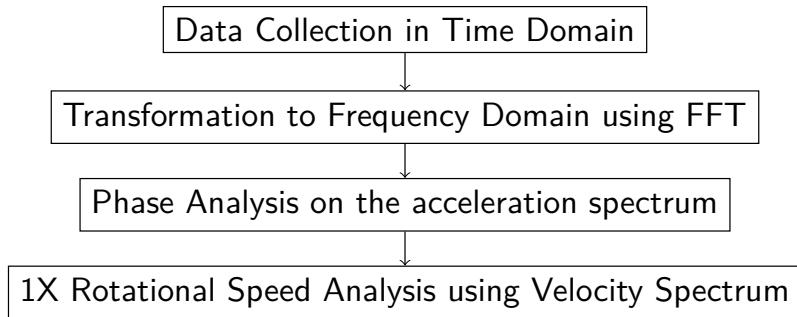


Figure 7: The series of checks carried out by the algorithm

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Incipient SVM Fault Detection Model and Anomaly Detection Model on time-domain dataset

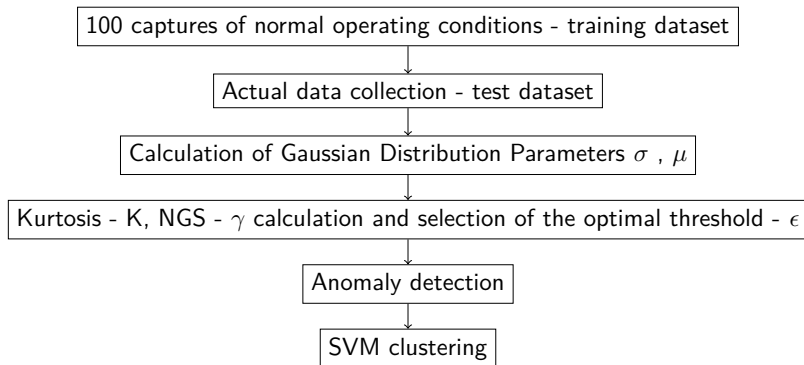


Figure 8: Incipient SVM Model

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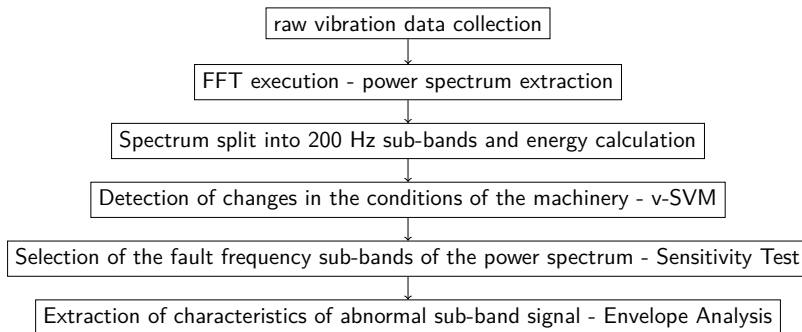


Figure 9: Architecture of anomaly detection frequency sub-band [?]

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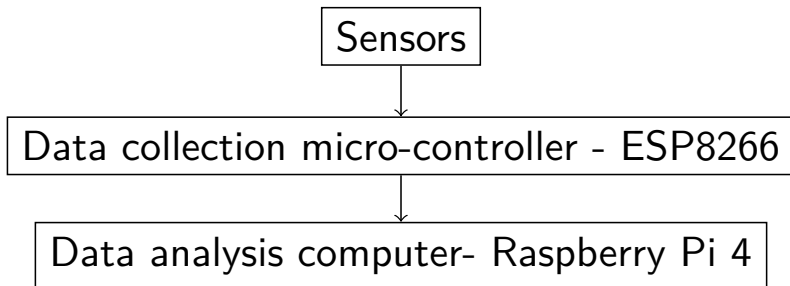


Figure 10: Architecture of anomaly detection frequency sub-band [?]

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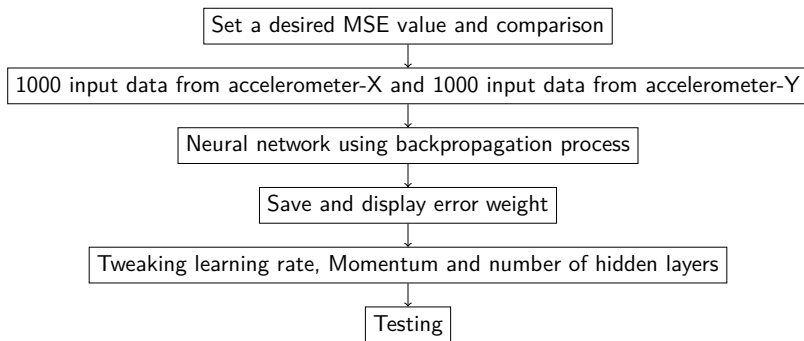


Figure 11: Flowchart for the Backpropagation-Neural Network (B-NN) training and testing [?].

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- **Vibration Analysis Techniques:** Wide variety including time-domain and frequency-domain analysis.
- **Fault Detection Algorithms:** Use parameters or thresholds for classification, highly effective for assessing machinery health. Dynamic signal stability analysis is a standout tool.
- **Machine Learning Anomaly Detection:** Can be combined with root algorithms, focus on continuous monitoring and past data analysis in specific application contexts.

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